

Responding to Ransomware Attacks with Microsoft Defender XDR

Microsoft Learn field guide for containment, investigation, eradication, and recovery

Condensed, non-verbatim operational guide

Purpose	Quick-reference guide for responding to suspected or active ransomware attacks using Microsoft Defender XDR incident evidence, containment actions, and recovery planning.
Primary audience	SecOps analysts, incident responders, Defender XDR administrators, identity and endpoint teams, and security leadership.
Reference source	https://learn.microsoft.com/en-us/defender-xdr/playbook-responding-ransomware-m365-defender
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Use note	This document is a condensed field guide based on Microsoft Learn guidance. Use the linked Microsoft Learn article as the authoritative source for the latest portal procedures, response actions, and product behavior.

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1. Response objective and expected outcome

The response objective is to quickly establish secure incident communications, assess ransomware scope, contain impacted entities, investigate root cause and attacker activity, and move toward eradication and recovery with a clear record of actions taken.

Microsoft Learn response framing

The Microsoft Learn playbook says that when ransomware is suspected or ongoing, secure communications with the incident response team should be established immediately. It organizes response into investigation and containment, followed by eradication and recovery.

2. At-a-glance response workflow

Step	Phase	Focus
1	Secure communications	Move incident coordination into a secure channel that is not dependent on potentially compromised services.
2	Assess scope	Use Microsoft Defender XDR incidents, alerts, and evidence to understand impacted or at-risk users, devices, apps, payloads, and network activity.
3	Contain fast	Containment and investigation should run together, but the playbook emphasizes quickly achieving containment so there is more time to investigate.
4	Preserve existing systems	Protect backups and existing systems while preventing further attacker spread or payload execution.
5	Investigate and eradicate	Investigate root cause, impacted assets, and attacker activity; then remove malicious artifacts, persistence, and attack paths.
6	Recover and monitor	Restore business function safely, validate recovered systems, monitor for recurrence, and document lessons learned.

3. Scope questions from the playbook

Microsoft Learn lists incident-scope questions that can be answered using the alerts and evidence in the Defender XDR incident. Use this table as the starting triage worksheet.

Scope area	Questions to answer
Compromised accounts	Which user accounts might be compromised? Which accounts were used to deliver the payload?
Affected devices	Which onboarded and discovered devices are affected, and how? Separate originating, impacted, and suspicious devices.
Network activity	Identify any network communication associated with the incident.
Applications and payloads	Which applications are affected? What payloads were spread? What was the payload delivery medium?
Command and control	How is the attacker communicating with compromised devices? If IP or URL information is available and Network Protection is enabled, the source references adding blocks through the indicators page.

4. Response checklist

Secure communications and incident setup

- ☐ Establish secure communications with the incident response team immediately.

- ☐ Define response leadership, decision logging, investigation ownership, and containment approval path.
- ☐ Preserve relevant alerts, evidence, logs, system context, and timeline notes.

Containment and scope assessment

- ☐ Use Microsoft Defender XDR to review impacted or at-risk assets in the incident.
- ☐ Identify compromised users, accounts used to deliver payloads, affected devices, suspicious devices, applications, payloads, and network communication.
- ☐ Prioritize containment while continuing investigation, because the source emphasizes containment and investigation occurring simultaneously with focus on fast containment.
- ☐ If indicators such as IPs or URLs are known and Network Protection is enabled, consider blocking them using the indicators workflow referenced by Microsoft Learn.

Preserve systems and prevent further damage

- ☐ If online backups exist, consider disconnecting the backup system from the network until confident the attack is contained.
- ☐ If active ransomware deployment is occurring or expected, suspend suspected privileged and local accounts according to approved procedures.
- ☐ Stop relevant remote logon sessions where appropriate.
- ☐ Reset compromised user account passwords and require users of compromised accounts to sign in again.

Investigation, eradication, and recovery

- ☐ Investigate payload delivery, lateral movement, affected applications, compromised accounts, and attacker communication paths.
- ☐ Remove malicious artifacts and persistence mechanisms after preserving evidence needed for investigation.
- ☐ Restore services safely only after containment and validation steps support recovery.
- ☐ Monitor restored systems and impacted accounts for recurring attacker activity.

Closure and improvement

- ☐ Document a complete timeline, containment actions, investigation findings, eradication steps, and recovery status.
- ☐ Capture detection and response gaps and convert them into follow-up work items.
- ☐ Review prevention guidance and related ransomware playbooks after the incident.

5. Detailed field guidance

5.1 Establish secure communications first

- The Microsoft Learn playbook explicitly says to establish secure communications with the incident response team immediately when ransomware is suspected or active.
- Use this secure channel to coordinate containment, evidence handling, executive updates, business impact decisions, and recovery planning.
- Keep a decision log so response choices are reviewable after the incident.

5.2 Containment and investigation should overlap

- Microsoft Learn states that containment and investigation should occur as simultaneously as possible.
- The same source says responders should focus on quickly achieving containment so there is more time to investigate.
- Use Defender XDR incident evidence to decide which entities require immediate isolation, account action, indicator blocking, or further investigation.

5.3 Use the incident evidence list to scope impact

- Microsoft Defender XDR can provide a consolidated view of impacted or at-risk assets to aid incident response assessment.
- Use the incident alerts and evidence list to identify compromised accounts, payload delivery accounts, originating devices, impacted devices, suspicious devices, network communication, affected applications, payloads spread, and payload delivery medium.
- Keep the triage worksheet updated because containment decisions should be grounded in the latest scope evidence.

5.4 Preserve systems and backups

- The source says that if online backups exist, consider disconnecting the backup system from the network until confident the attack is contained.
- Backup preservation should be coordinated with infrastructure and business continuity teams so recovery options are not accidentally destroyed.
- Validate that restored systems are not reconnected broadly until containment and monitoring support safe recovery.

5.5 Move from response to recovery

- The playbook organizes later response into eradication and recovery after investigation and containment.
- Eradication should address malicious artifacts, compromised accounts, persistence mechanisms, and attacker communication paths identified during investigation.
- Recovery should restore business operations in a controlled sequence with validation and post-recovery monitoring.

6. Evidence and decision matrix

Question	Evidence / decision input
Is this an active ransomware incident?	Defender XDR incident, alerts, evidence list, endpoint activity, payload evidence, and business-impact reports
Which accounts are involved?	Potentially compromised users, accounts used to deliver payload, privileged/local accounts suspected as part of the attack

Which devices are affected?	Originating devices, impacted devices, suspicious devices, onboarded and discovered devices
How is the attacker communicating?	Network communication in the incident, IP/URL indicators where available, Network Protection context
What payloads and apps are involved?	Payloads spread, affected applications, payload delivery medium, associated alerts and evidence
What should be contained first?	Entities most likely enabling active spread, payload deployment, or privileged control
What can recover safely?	Backup status, containment confidence, restored system validation, monitoring for recurrence

7. Closure notes template

- ☐ Incident summary: suspected or confirmed ransomware activity, detection source, business impact, and current status.
- ☐ Timeline: first alert, secure communications established, containment actions, investigation milestones, eradication, and recovery completion.
- ☐ Scope: compromised accounts, delivery accounts, originating devices, impacted devices, suspicious devices, affected apps, payloads, and network communication.
- ☐ Evidence reviewed: Defender XDR incident, alerts, evidence list, endpoint and identity telemetry, network indicators, backup status, and recovery validation.
- ☐ Containment actions: accounts suspended or reset, sessions stopped, indicators blocked, devices isolated, backups protected, and remote access reduced where applicable.
- ☐ Eradication and recovery: malicious artifacts removed, persistence addressed, systems restored, users re-enabled, and monitoring confirmed.
- ☐ Follow-up recommendations: prevention guidance review, detection tuning, response playbook updates, tabletop exercise, and ownership tracking for remediation gaps.

8. Authoritative sources

Primary Microsoft Learn article: [Responding to ransomware attacks | Microsoft Defender XDR | Microsoft Learn](#)

Related detection playbook: [Detecting human-operated ransomware attacks with Microsoft Defender XDR | Microsoft Learn](#)

Related Defender XDR overview: [What is Microsoft Defender XDR? | Microsoft Learn](#)

Related attack disruption overview: [Automatic attack disruption in Microsoft Defender | Microsoft Learn](#)

Related ransomware hub: [Microsoft defense against ransomware, extortion, and intrusion | Microsoft Learn](#)

This field guide does not replace the official articles. Always reference Microsoft Learn for the latest product behavior, portal guidance, and response recommendations.